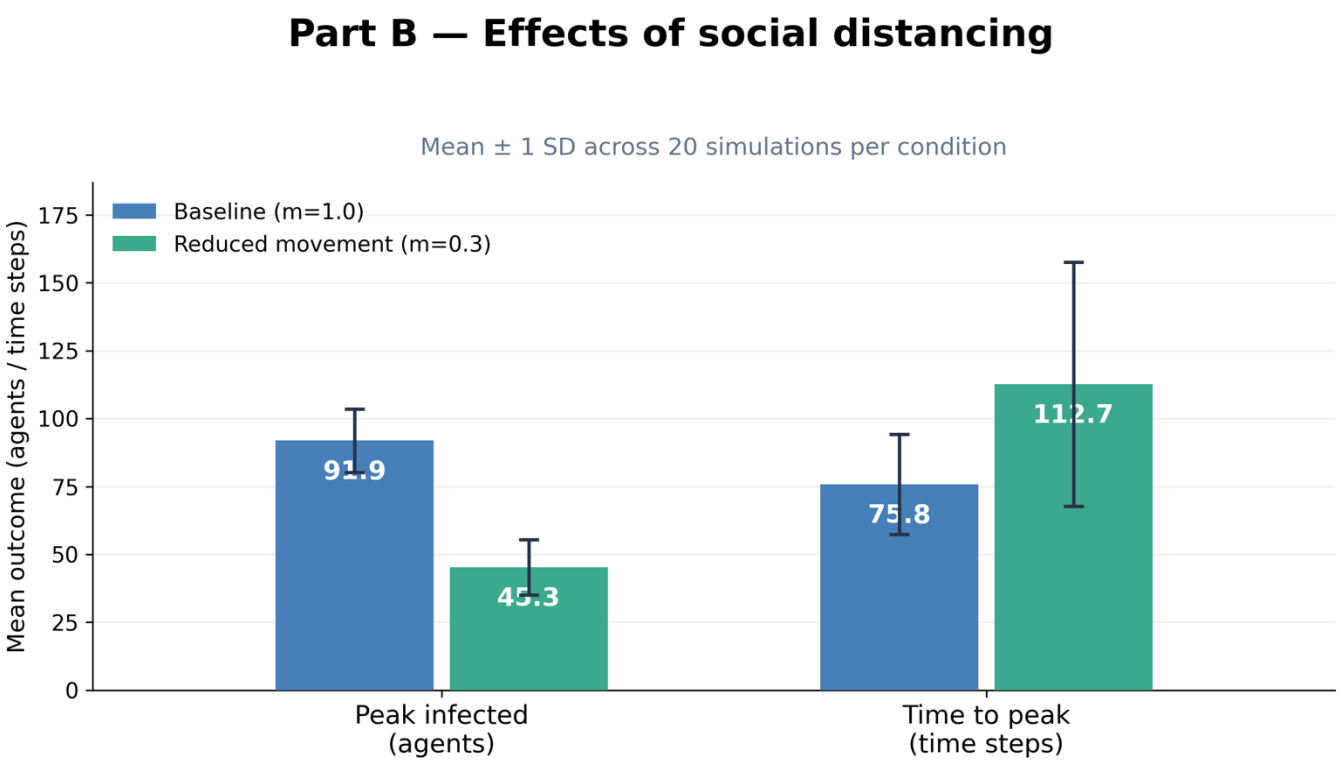
Less movement, smaller outbreaks

Agent-based model of pandemic spread · Lige Zhang · BMI 500, HW 2 in Lecture 7



200 agents (5 initially infected) on a 20 × 20 grid; p = 0.30, q = 0.02; 300 steps. Means of per-run outcomes over 20 seeds. Blue = baseline (m = 1.0); Green = reduced movement (m = 0.3).

MOSTLY SUPPORTED

HYPOTHESIS

Reducing agents' movement will lower the infection peak and delay it.

The average infection peak falls when reducing agents' movement; on average, the peak is also delayed, but there are a few rare cases that the peak comes early.

KEY NUMBER

91.9 → 45.3

Mean peak infected drops significantly!

LESSON ABOUT MODELING

One stochastic run is not a result.

Identical settings can induce very different results, so we need multiple runs to solidify the findings.